Jacopo Di Silvestre

Disease Simulation Modeling and Cost-Effectiveness Analyses

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linkedin.com/in/jacopo-di-silvestre | github.com/jacopo-disilvestre/LCMicrosim

Professional Summary

Health economics researcher and PhD candidate who builds and works on microsimulation models in C++ and R to evaluate cancer screening strategies and their cost-effectiveness. Experience includes individual-based model development and recalibration of the CISNET Lung Cancer Policy Model (LCPM). Research spans lung and cervical cancer screening, smoking trends, and HPV vaccination evidence review.

Technical Skills

Programming: C++, R, Bash

Computing and workflow: Git, Slurm, High-performance computing (HPC), Visual Studio

Analysis and reporting tools: Excel, LaTeX, Microsoft Office 365

Research Experience

PhD Candidate Mar 2025 - Present

University of Oslo, Oslo, Norway

- Conduct research within the Kreftforeningen-funded LC-CONTROL project on lung cancer screening in Norway.
- First author of a study of age- and sex-specific smoking trends by birth cohort in Norway, using cross-sectional survey data from 1978-2024.
- Research topics include screening eligibility criteria and two-stage risk-based approaches incorporating SNP-assisted risk prediction.

Research Scholar Nov 2024 - Present

Harvard Medical School - Massachusetts General Hospital
Center for Health Technology Assessment (MGH-CHTA), Boston, MA

- Recalibration of the Lung Cancer Policy Model (LCPM), a CISNET microsimulation model, to support analyses in the United States and Norway.
- Apply simulation modeling and C++ programming in lung cancer research.
- On-site research visits: Nov 2024 - Jan 2025; Aug - Sep 2025; Oct 2026.

Research Assistant Jul 2024 - Feb 2025

University of Oslo, Oslo, Norway

- Developed an individual-based lung cancer natural history model in R to evaluate the cost-effectiveness of risk-based low-dose CT screening in a high-risk Norwegian population.
- Analyzed outputs from the CISNET Harvard-CC model in R for a cost-effectiveness study of cervical cancer screening strategies in Norway.
- Conducted a systematic review of the health impact and cost-effectiveness of one-dose human papillomavirus (HPV) vaccination compared with alternative schedules.

Research Experience (continued)

Research Intern Sep 2023 - Feb 2024
ASCERTAIN European Project

- Conducted a scoping review of statistical methods for estimating comparative clinical effectiveness using evidence from single-arm or non-randomized trials.

Education

PhD Studies Mar 2025 - Present
University of Oslo, Norway

MS in Health Economics and Health Management Sep 2022 - Jun 2024
Erasmus University Rotterdam

- European Master in Health Economics and Management (EU-HEM), with study at the University of Oslo and University of Bologna.
- **Thesis:** "Risk-Based Lung Cancer Screening: A Microsimulation Model for the Norwegian High-Risk Population."
- The thesis analysis estimated NOK 129,689 per quality-adjusted life-year (QALY) gained for the evaluated risk-based screening strategy versus no screening in the modeled high-risk Norwegian population.
- **Relevant coursework:** Advanced Health Economic Modeling; Health Technology Assessment; Pharmaceutical Pricing and Market Access; Econometrics; Discrete Choice Experiments.

BS in Economics 2019 - 2022
University of Pavia, Italy
Grade: 109/110.

Selected Publications

Di Silvestre J, Salgado MV, Jeon J, Mok Y, Holford TR, Vedøy TF, Pedersen K, Burger EA, Meza R. Age- and sex-specific smoking trends by birth cohort in Norway, 1978-2024: an analysis of cross-sectional survey data. *The Lancet Regional Health - Europe*. 2026;71:101886.
doi: 10.1016/j.lanepe.2026.101886

Pedersen K, **Di Silvestre J**, Sy S, Portnoy A, Castle PE, Kim JJ, Burger EA. Optimizing Cervical Cancer Screening by Age at Vaccination for Human Papillomavirus: Health and Resource Implications. *Annals of Internal Medicine*. 2026.
doi: 10.7326/ANNALS-25-03192

Teaching and Supervision

Supervise master's students on thesis model design and programming.

Selected Training

NRIS HPC On-Boarding Course.